

Multiple Choice (10 marks)

1. What is leptin resistance?
 - (a) Deficiency of leptin caused by a mutation in the leptin gene
 - (b) Lack of leptin release from the adipose tissue
 - (c) High concentrations of leptin without reduction in energy intake
 - (d) Low concentrations of leptin with increased energy intake
2. Which of the following statements is/are true? How can smoking affect breastfeeding?
 - (a) Suppresses milk production
 - (b) Alters the composition of breast milk
 - (c) Increases the risk of early cessation of breastfeeding
 - (d) all of the options given are correct
3. What is the best method to estimate changes in body fat during weight loss?
 - (a) BMI
 - (b) DXA
 - (c) Four-compartment model
 - (d) MRI
4. Which of the following confirmed values meet the diagnostic threshold for diabetes?
 - (a) fasting blood glucose $\geq$ 140 mg/dl
 - (b) random glucose $>$ 160 mg/dl
 - (c) 2 hour post prandial glucose $\geq$ to 126 mg/dl
 - (d) fasting blood glucose $\geq$ 126 mg/dl
5. Which of the following vitamins is involved in one-carbon metabolism?
 - (a) Folate
 - (b) Riboflavin
 - (c) Thiamin
 - (d) Vitamin C

True / False (10 marks)

Answer True or False

6. True or False: Acetaldehyde is a significant metabolite of alcohol that contributes to cancer development.
7. True or False: Transaminases require a coenzyme derived from vitamin B 6 for the process of transamination.
8. True or False: Carbon 13 is frequently employed for measuring total body water in medical studies.
9. True or False: Food availability and consumption can be measured across five distinct stages within the food supply chain.
10. True or False: Low intake of fruits and vegetables is a primary cause of iodine deficiency in all regions of the world.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss how alcohol consumption can impact the risk of developing oesophageal cancer, including the biological mechanisms involved and potential preventative measures.
12. Discuss the role of Vitamin B 6 in the process of transamination of amino acids, including its significance in metabolism and overall health.

End of Paper